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Trace Dual-Crosslinkable Additives Enable Direct Microlithography for Enhanced Organic Electrochemical Transistors

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Similar to silicon-based electronics, the implementation of micro/nano-patterning to facilitate complex device architectures and high-density integration is crucial to the development of organic electronics. Among various patterning techniques, direct microlithography (DML) is highly applicable and extensively adopted in organic electronics, such as organic electrochemical transistors (OECTs). However, conventional DML often requires high crosslinker concentrations, leading to compromised electrical performance. To address this challenge, a novel strategy is developed that combines supramolecular and covalent interactions by incorporating a polyrotaxane supramolecular crosslinker (PR) into poly(benzodifurandione) (PBFDO). The PR forms a hydrogen bonding network with PBFDO and undergoes UV-triggered covalent crosslinking among its molecules, providing solvent resistance even at trace loading levels (<0.1 wt%). This approach enables precise patterning of PBFDO with feature sizes below 1 µm while preserving high electrical performance. Notably, PR also serves as a performance enhancer, promoting molecular ordering and ionic conduction within PBFDO. OECTs fabricated with PR-crosslinked PBFDO exhibit about one-order-of-magnitude increase in ON/OFF ratio, a 42% increase in μC^* (reaching 2460 F cm^{-1} V^{-1} s^{-1}), and elevated operational stability compared to pristine ones. This multifunctional crosslinker offers a scalable solution for high-performance, high-density organic electronics and opens new avenues for supramolecular chemistry applications in this field.

1. Introduction

As in silicon-based electronics, the application of micro/nano-patterning is crucial to advancing organic electronics by enabling complex device structures and high-density integration.[1,2] Precise patterning allows for device miniaturization, minimizes parasitic capacitance, and facilitates the integration of complex circuits, significantly enhancing overall device performance and functionality.[3,4] Photolithography stands out as the premier technique for achieving such precise patterning due to unparalleled resolution and alignment accuracy. However, applying photolithography directly to organic semiconductors (OSCs) poses significant challenges. Conventional photolithographic processes often involve solvents that lack chemical orthogonality with the active OSC layers, leading to compromised device performance.[5,6] Recently, photo-assisted direct microlithography (DML) methods have been devised as an effective way for OSCs patterning, wherein crosslinkers are incorporated

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DOI: 10.1002/adma.202417452

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into OSCs to form insoluble networks upon UV exposure, thereby enabling precise patterning.[7–10] These methods show strong applicability and have been widely adopted in devices such as organic light-emitting diodes,[11] organic field effect transistors,[12] and organic electrochemical transistors (OECTs),[13] paving the way for advanced organic electronic technologies.

However, current DML strategies often require high loading levels of crosslinkers to achieve sufficient crosslinking density for effective patterning.[14–16] Whether crosslinking occurs through reactions with the OSCs or among crosslinker molecules themselves, the excessive concentration of crosslinker introduces insulators into the active layer. This often leads to molecular disorder, reduced carrier mobility, and introduces defects that deteriorate the overall electrical performance within the semiconductor matrix.[1,8,14,17] Moreover, when covalent crosslinking involves reactions with the conjugated polymer, it can disrupt the π-conjugation pathways essential for charge transport, especially when cross-linked side chains interpenetrate with the conjugated backbones, further exacerbating the decline in electrical performance.[7,8,15,18,19] Consequently, the trade-off between achieving high pattern ability and maintaining optimal electrical properties poses a substantial challenge in developing DML for OSCs.

In this work, recognizing that crosslinkers remain embedded within the OSC during DML, we sought to repurpose the crosslinker beyond its role as a mere patterning agent. By designing a crosslinker capable of efficient crosslinking at trace loading levels while enhancing the material's performance through intermolecular interactions, we addressed the limitation of high crosslinker concentrations and associated performance degradation. Specifically, we developed a dual crosslinking strategy that combines supramolecular and covalent interactions, using a polyrotaxane supramolecular crosslinker (PR) incorporated into the n-type organic mixed ionic-electronic conductors (OMIECs) poly(benzodifurandione) (PBFDO). The PRs are capable of undergoing UV-induced covalent crosslinking with one another, and the hydroxyl groups on the PRs can form hydrogen bonds (H-bonds) with multiple PBFDO chains, thereby facilitating the formation of a supramolecular crosslinking network that enhances the solvent resistance of the PBFDO film, even at minimal additive concentrations. This dual crosslinking network endows the system with the solvent resistance required for photolithography (**Figure 1a**). As a result, we achieved precise patterning of PBFDO films with feature sizes of 2 µm at trace PR loading levels (<0.1 wt%), and large-area patterning is also demonstrated with excellent homogeneity (Figure 1b,c). To the best of our knowledge, this represents the lowest crosslinker loading level reported for DML (Figure 1d, Table S1, Supporting Information). With a higher PR addition, higher resolution (≈600 nm) can be achieved, which is limited by the inherent resolution of the mask (Figures S9–S11, Supporting Information). The successful crosslinking of the film is evidenced by the drastically increased film retention ratio upon the addition of different mass ratio PR to PBFDO (Figure 1e; Figure S12, Table S2, Supporting Information).[7,20] Atomic force microscopy (AFM) further demonstrates the uniformity of the PBFDO:PR blend film (Figure S13, Table S3, Supporting Information), ensuring that the crosslinking network forms evenly across the entire film.

Notably, our dual crosslinking system maintains consistent electrical performance across a wide range of crosslinker loading levels compared to conventional covalent-based crosslinking methods reported in the literature (Figure S14, Supporting Information).[7,21] This can be attributed to the non-covalent interactions that maintain the integrity of the π-conjugation pathways in PBFDO. Additionally, by leveraging these intermolecular interactions, the PR promotes better molecular order and optimizes ion conduction and coupling within PBFDO. This improvement is exemplified in OECTs, where devices with the crosslinked active layer exhibit about one-order-of-magnitude increase in ON/OFF ratio, a 42% increase in the product of charge-carrier mobility and volumetric capacitance ($\mu C^* = 2460$ F cm^{-1} V^{-1} s^{-1}), and enhanced operational stability compared to devices based on pristine PBFDO. Furthermore, a high-density device array (2.2×10^5 devices cm^{-2}) was demonstrated with good performance uniformity.

Overall, the crosslinker acts both as a patterning agent and a performance enhancer, addressing the critical challenge of balancing patternability and electronic properties in OSCs in photolithography. Our approach provides a scalable solution for fabricating high-performance, high-density organic electronic devices and opens new avenues for the application of supramolecular chemistry in the field.

2. Results and Discussion

2.1. Mechanism of Dual Crosslinking Strategy

The PR synthesized in this work consists of a polyethylene glycol (PEG) backbone that threads through several cyclodextrin (CD) rings, with both ends capped by bulky adamantane groups. PEG enhances the aqueous solubility of PR, while the hydroxyl groups on CD serve as H-bond donors to interact with PBFDO (**Figure 2a**). Additionally, several hydroxyl groups on the CD units are functionalized with photoreactive poly(ethylene glycol) methacrylate (PEGMA) side chains, which introduce covalent crosslinking upon UV exposure (Figure 2b). ^{1}H NMR spectra illustrate the components that compose a PR molecule (Figures S1–S8, Supporting Information), showing an average of 9–10 CD molecules threaded onto the PEG backbone, with each PR molecule containing an average of 5–6 PEGMA side chains (Figures S5 and S8, Supporting Information). The design of this photo-cross-linkable molecular structure provides the foundation for achieving DML under an ultra-low concentration.

Raman spectroscopy was applied to confirm the formation of the supramolecular crosslinking network via H-bonds interactions between PR and PBFDO. The C═C stretching at 1509 cm^{-1} and C—H in-plane bending vibrations at 1259 cm^{-1} in PBFDO,[24] characteristic of its conjugated backbone, exhibit little shift with the addition of PR (Figure S15, Supporting Information), suggesting that there are no additional interactions between the PBFDO backbone and PR. Moreover, noticeable redshifts were observed in the Raman peaks corresponding to C═O stretching (shifting from ≈1621 to 1617 cm^{-1}) (Figure 2c) and C—O—C symmetric stretching vibrations (shifting from ≈1107 to 1100 cm^{-1}) (Figure 2d).[25] The redshifts can be attributed to the weakening of the C═O and C—O—C bonds due to the formation of hydrogen bonding that reduces the overall vibrational energy. In addition,

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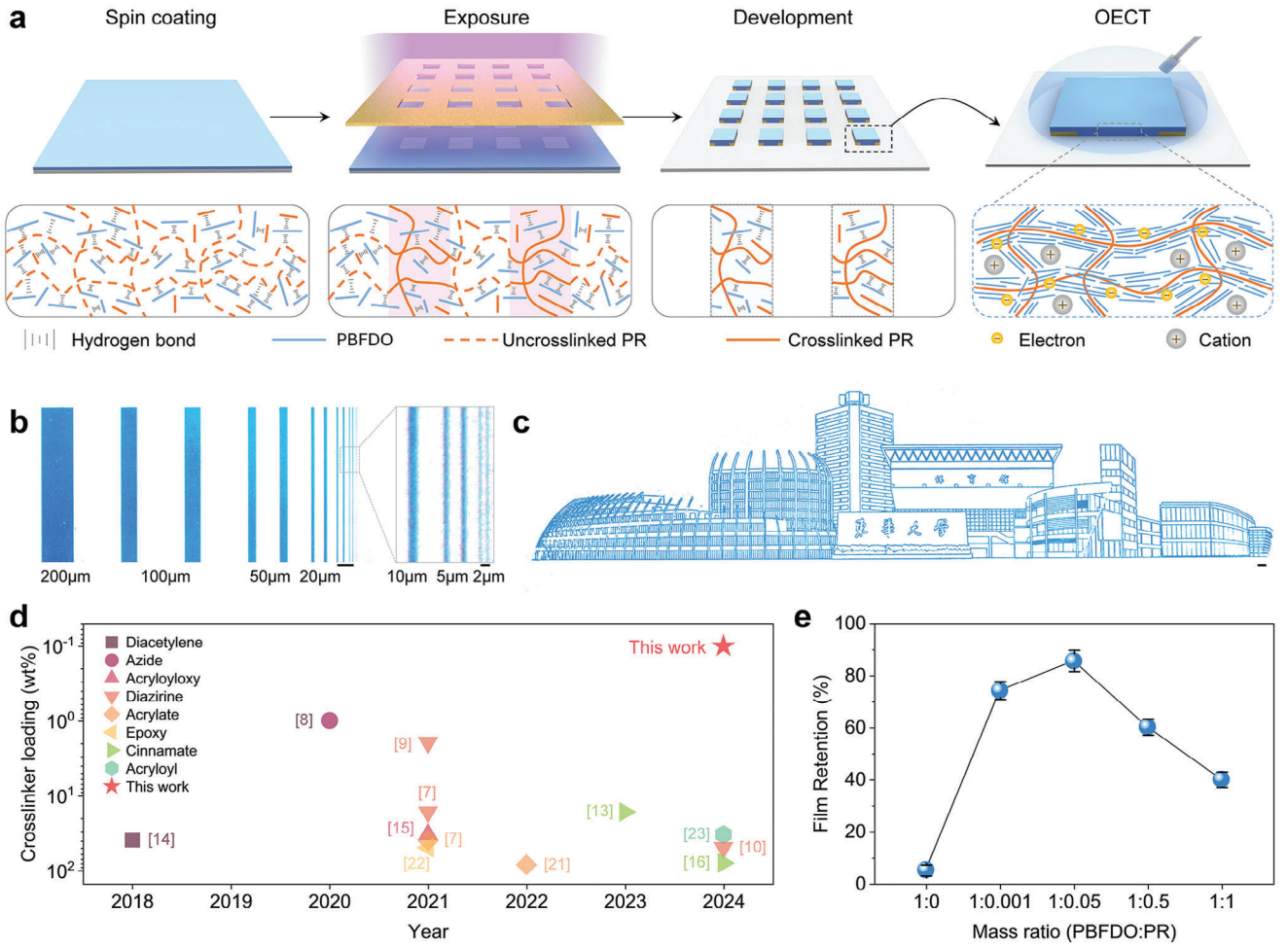


Figure 1. a) DML process based on dual crosslinking network strategy. b) Optical microscopy images of PBFDO:PR = 1:0.001 film, demonstrating resolution down to 2 μm (left scale bar: 100 μm; right scale bar: 10 μm). c) Image of intricate patterns (scale bar: 300 μm). d) The lowest reported crosslinker loading (<0.1 wt%) for OMIECs patterning to date.[7–10,13–16,21–23] e) Thickness retention ratios of pristine PBFDO film and PBFDO:PR = 1:0.001/0.05/0.5/1 films.

the slight red shift observed at the 1593 cm^{-1} peak suggests that when C═O or C—O—C groups on the PR form H-bonds with the hydroxyl groups on PBFDO, the electron density or electronic interactions of the molecule are altered. To confirm the formation of the covalent crosslinking network, infrared spectroscopy was performed on the mixture of photo-functionalized side-chain PEGMA and 2-hydroxy-2-methylpropiophenone (photoinitiator 1173). As a result, with the light exposure time increasing, the C═C stretching vibration at 1637 cm^{-1} gradually declines, indicating a covalent crosslinking network forms that derives from PEGMA (Figure 2e).[26]

Density functional theory (DFT) was employed to calculate the binding energies of different types of H-bonds between PBFDO and PR. The binding energies are 27.45 kJ mol^{-1} for $O—H\cdots O_{(C=O)}$ and 4.31 kJ mol^{-1} for $O—H\cdots O_{(C—O—C)}$ (Figure 2f; Figure S16, Supporting Information). All-atom molecular dynamics (MD) simulations were conducted to model the PBFDO system (Figure 2g). The simulation consists of 10 random PR polymer chain segments (each represents 10% of a PR molecule) and 300 PBFDO polymer chain segments (the number of repeating units (n) is 8, representing 2% of a PBFDO molecule) (Figure S17, Supporting Information). The relationship between polymer chain segments and polymer chains is derived from the molecular weight measured by gel permeation chromatography (GPC) (Table S4, Supporting Information). These polymer chains were equilibrated within the simulation cell to replicate the conformations of real polymer chains at equilibrium. The computational results reveal that ≈45 H-bonds appear under standard temperature and pressure conditions (Figure 2h), corresponding to ≈45 H-bonds forming when one PR molecule interacts with 6 PBFDO molecules, meaning that one PR molecule forms ≈7 to 8 H-bonds with each PBFDO molecule. The radial distribution function shows a primary peak at 0.2–0.3 nm, indicating a high probability of H-bond formation within this distance range (Figure S18, Supporting Information). These results demonstrate the efficiency of H-bond formation between PR and PBFDO.

Based on DFT calculations, we assume all H-bonds form specifically between hydroxyl groups and furanone carbonyl sites, resulting in a total hydrogen bonding energy of

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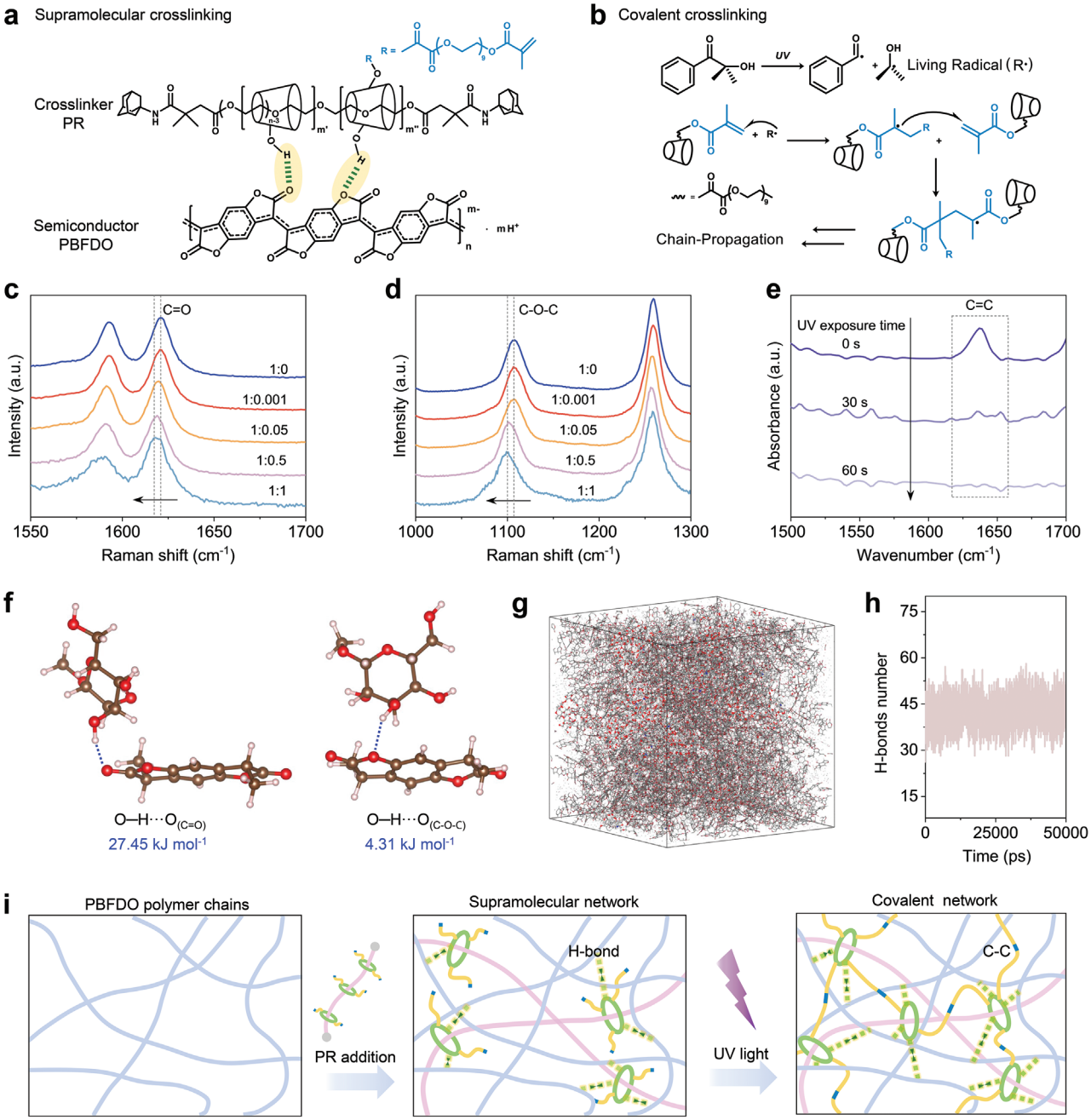


Figure 2. a) Supramolecular crosslinking: H-bonds between PBFDO and PR. b) Covalent crosslinking: chain-growth polymerization. c) Raman spectra of pristine PBFDO film and PBFDO:PR films with different mass ratios in the 1550–1700 cm^{-1} range. d) Raman spectra of pristine PBFDO film and PBFDO:PR films with different mass ratios in the 1000–1300 cm^{-1} range. e) The infrared spectrum shows that the C═C signal gradually disappears with the increase of UV illumination time. f) The binding energy of H-bonds ($O—H\cdots O_{(C=O)}$ and $O—H\cdots O_{(C—O—C)}$) was calculated from DFT. g) Snapshots for all-atom MD simulations of the PBFDO:PR structures. h) The number of H-bonds presenting in the PBFDO:PR system was calculated from MD simulations. i) Dual-network crosslinking mechanism of the supramolecular crosslinker PR in OMIEC: H-bond crosslinking and covalent crosslinking through functionalized C═C bonds.

≈1235 kJ mol^{-1} in this system. Although an individual H-bond is relatively weak, the presence of multiple H-bonds in the PBFDO:PR blended system ensures the formation of a robust H-bond crosslinking network. Additionally, the synthesized PR molecules were functionalized with up to 6 PEGMA chemical crosslinking side chains. After UV-triggered crosslinking, the bond energy of each resulting C—C bond is ≈345 kJ mol^{-1}.[27] If all functionalized end groups react, the total bond energy from covalent crosslinking would amount to ≈2070 kJ mol^{-1}. This result strongly indicates that even at very low PR loading, the combined intermolecular forces significantly contribute to network formation.

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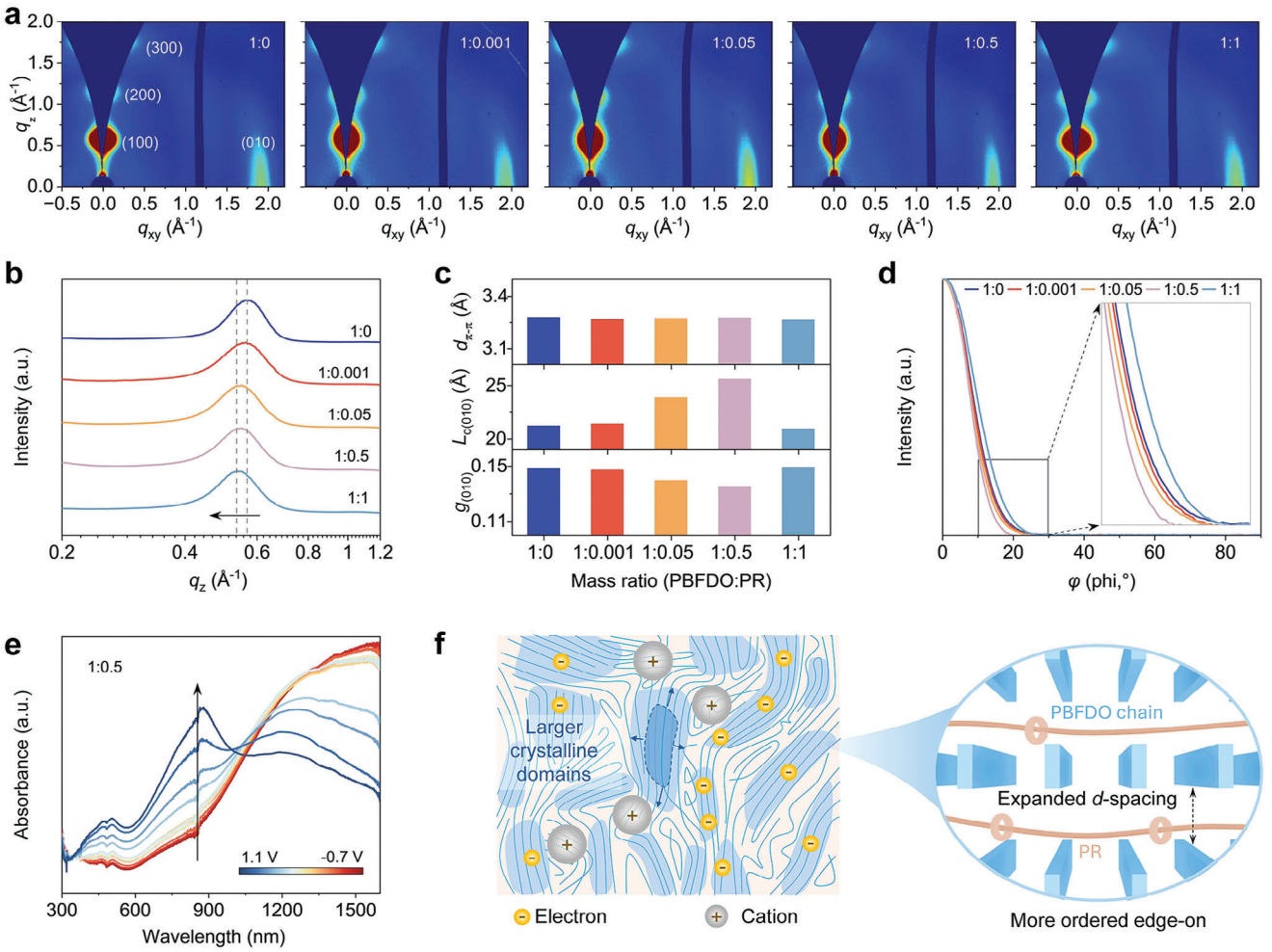


Figure 3. a) 2D GIWAXS patterns of pristine PBFDO film and PBFDO:PR = 1:0.001/0.05/0.5/1 films. b) Out-of-plane line cuts show the shift of 100 scattering peaks. c) Variation trends in $d_{\pi-\pi}$, $L_{c(010)}$, and $g_{(010)}$ for the pristine PBFDO film and PBFDO:PR = 1:0.001/0.05/0.5/1 films. d) Angular integration of the 100 peaks in the range of 0.3–0.6 for pristine PBFDO film and PBFDO:PR films with different mass ratios. e) Normalized in-situ electrochemical UV–vis absorbance of PBFDO:PR = 1:0.5 film. f) Schematic of microstructure and ion doping.

Therefore, the dual crosslinking mechanism enables precise patterning at low PR concentrations, demonstrating the effectiveness of the PR design in leveraging both intermolecular noncovalent bond forces and covalent bonding to achieve efficient patterning with minimal material addition. The supramolecular crosslinking occurs through H-bond interactions between the hydroxyl groups on the CD units and the PBFDO polymer chains, allowing one PR molecule to stabilize multiple PBFDO chains. Upon UV exposure, the PEGMA side chains undergo photoinduced polymerization, forming covalent bonds that further reinforce the network, providing the solvent resistance necessary for microlithography (Figure 2i).

2.2. PR-Induced Structural Changes

PR also acts as a morphology-modulating additive in PBFDO film to provide a balanced ion and electron charge transport. As shown in the result of grazing incidence wide-angle X-ray scattering (GIWAXS), all films exhibit similar crystal texture (**Figure 3a**). The pristine PBFDO film exhibits a strong lamellar stacking (100) scattering peak at $q_z = 0.57$ Å^{-1} and a π–π stacking (010) scattering peak at $q_{xy} = 1.92$ Å^{-1}, indicating a d-spacing of 11.10 Å and π–π stacking distance of 3.28 Å, respectively (Tables S5, S6, Supporting Information).[28] Upon the addition of PR, the lamellar stacking scattering peak shifts to $q_z = 0.54$ Å^{-1} (Figure 3b), corresponding to an increased d-spacing of 11.58 Å at 1:0.5 mass ratio (Figure S19, Supporting Information). The expansion in d-spacing suggests that the PR molecules are intercalating between the PBFDO chains, pushing PBFDO further apart while maintaining the overall crystalline structure. Interestingly, the π–π stacking distance remains largely unaffected by the addition of PR (Figure 3c), which indicates that the conjugated backbone of PBFDO is not disrupted by the crosslinking process. The crystal coherence length (L_c) and paracrystalline disorder (g) of the π–π stacking (010) scattering reveal an initial increase in L_c, followed by a decrease at higher PR concentrations, with the maximum L_c observed at a PBFDO:PR mass ratio of 1:0.5 (Figure 3c,

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Table S5, Supporting Information). Conversely, at higher PR concentrations, excessive crosslinking can introduce defects that reduce the overall crystallinity of the material. The *g* parameter follows a similar trend, with minimal disorder observed at the PBFDO:PR mass ratio of 1:0.5. Hermans' orientation function (HF) shows an obvious increase from 0.942 in pristine PBFDO film to 0.986 in the PBFDO:PR = 1:0.5 film (Figure 3d, Table S7, Supporting Information), which presents an enhanced edge-on orientation with better order attributed to appropriate dosage of PR.[29]

In situ electrochemical UV–vis absorption spectroscopy was applied to investigate the effect of different mass ratios of PR crosslinker on the electrochemical doping and dedoping in PBFDO:PR films. Upon applying a potential of 1.1 V versus Ag/AgCl, the pristine PBFDO film exhibits two significantly enhanced absorption features: a high-energy band ranging from 450 to 550 nm (corresponding to the π–π^* transition) and a broad absorption band between 650 and 950 nm (indicative of intramolecular charge transfer [ICT] phenomena).[30] As the potential decreased, these absorption bands gradually weakened, accompanied by the emergence of new polaron absorption bands in the 950–1200 nm and 300–400 nm wavelength ranges (Figure S20a, Supporting Information).[30] For the PBFDO:PR film at a mass ratio of 1:0.5, the ICT absorption band increases significantly upon dedoping at 1.1 V (Figure 3e). Similar enhancement effects occur at other blending ratios (0.001 and 0.05) (Figure S20b,c, Supporting Information). However, when the ratio increases to PBFDO:PR = 1:1, the strength of the ICT absorption band drops at the dedoping voltage (Figure S20d, Supporting Information). Therefore, an optimal balance between crosslinking density and doping performance appears under the PR mass ratio of 1:0.5.

The results mentioned above suggest that the appropriate addition of PR leads to a well-ordered edge-on orientation in the films and presents larger crystalline regions, which facilitates electron transport,[31,32] moreover, the addition of PR increases the *d*-spacing of the layers, which aids in the movement of ions between the polymer chains, enhancing the overall electrochemical performance (Figure 3f; Figure S21, Supporting Information).

2.3. Enhanced Performance of PBFDO:PR Based OECTs

The ion-electron coupling transport performance of PBFDO:PR films is further evaluated in OECTs. These devices operate in 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ($[EMIM]^+$ $[TFSI]^-$), using Ag/AgCl pellets as the gate electrode in ambient air. Since all films possess high inherent conductivity (Figure S14, Supporting Information), all devices operate in depletion mode (Figure S22, Supporting Information). A large channel (500 μm × 500 μm) was used to obtain slow doping speeds, facilitating the comparison of dedoping rates for different films. As shown in **Figure 4a**, the turn-off delay time exhibits an obvious downward trend with the concentration of PR increasing. Furthermore, the maximum normalized transconductance ($g_{m,norm}$) of OECTs based on PBFDO:PR = 1:0.5 film (1421 S cm^{-1}) increased by ≈40% compared to pristine PBFDO OECTs (1016 S cm^{-1}) (Figure 4b). Notably, even a minimal photocrosslinker addition contributes to the enhancement of maximum normalized transconductance (from 1016 to 1056 S cm^{-1}) and significantly boosts the ON/OFF ratio (Figure 4c) and response time (Figure S23, Supporting Information). **Table 1** and Figure S24, S25 (Supporting Information) show the summary of electrical performance results for OECTs with different PR loading. At trace loading levels (PBFDO:PR = 1:0.001), μC^* increases from 1730 to 1860 F cm^{-1} V^{-1} s^{-1}, indicating an improvement in ion-to-electron transduction capability. For other PR blending mass ratios (0.05, 0.5, 1), the μC^* values are 2180, 2460, and 1510 F cm^{-1} V^{-1} s^{-1}, respectively, showing an initial increase followed by a decrease at higher blending ratios, with the peak μC^* value observed at the PBFDO:PR mass ratio of 1:0.5 (Figure S26, Supporting Information). This represents an ≈42% enhancement compared to pristine PBFDO and surpasses the highest μC^* values reported for n-type OECT devices (Figure 4d and Table S8, Supporting Information). By measuring the CV curves of films with different mass ratios at varying scan rates (Figures S27–S31, Supporting Information), the volumetric capacitance (C^*) initially increases from 139.65 F cm^{-3} (pristine PBFDO) to 191.14 F cm^{-3} (PBFDO:PR = 1:0.5) and decreases to 105.64 F cm^{-3} when PBFDO:PR = 1:1 (Figure S32, Supporting Information). Simultaneously, the calculated carrier mobility (μ) ranges from 12.39 to 14.29 cm^2 V^{-1} s^{-1} (Figure S33, Supporting Information).

The long-term stability of OECTs with different mass ratios of PR was evaluated. During 5000 pulse cycles, the films with added PR exhibit slower current decay compared to that of the pristine PBFDO (Figure 4e; Figure S34, Supporting Information). The cycling stability of the films is closely linked to the passive swelling (non-charging) and active swelling (during charging) that occur during the doping and dedoping process.[33–35] To quantify the expansion and contraction of the films during electrochemical operation, we employed electrochemical quartz crystal microbalance with dissipation monitoring (EQCM-D). Upon the incorporation of PR, the films exhibit a reduction in both passive and active swelling (Figures S35–S46, Table S9, Supporting Information), which effectively mitigates the risk of film breakdown and enhances cycling stability. However, when the amount of PR increases to 1:1, the excessive expansion of the interlayer distance leads to rapid ion insertion, causing an increased influx and efflux of hydrated molecules in the films that exacerbate swelling (Figure 4f).

To validate the potential of PBFDO:PR for organic integrated circuits, a fully photolithographic OECT array with a PBFDO:PR was fabricated to demonstrate monolithic integration, achieving an OECT density of 2.2 × 10^5 cm^{-2} (Figure 4g,h). A 1000 × 1000 PBFDO OECT array (1 M OECTs) was fabricated on a 2-inch wafer, with transfer characteristics measured from 64 uniformly selected devices across the entire area using an external Ag/AgCl electrode to assess reliability (Figure S47, Supporting Information). The devices exhibit excellent uniformity, with an average g_m of 1.22 ± 0.03 mS, highlighting the superior device consistency achieved through monolithic DML using PR.

3. Conclusion

In this work, we proposed a high-precision DML strategy that leverages the unique properties of a supramolecular crosslinker

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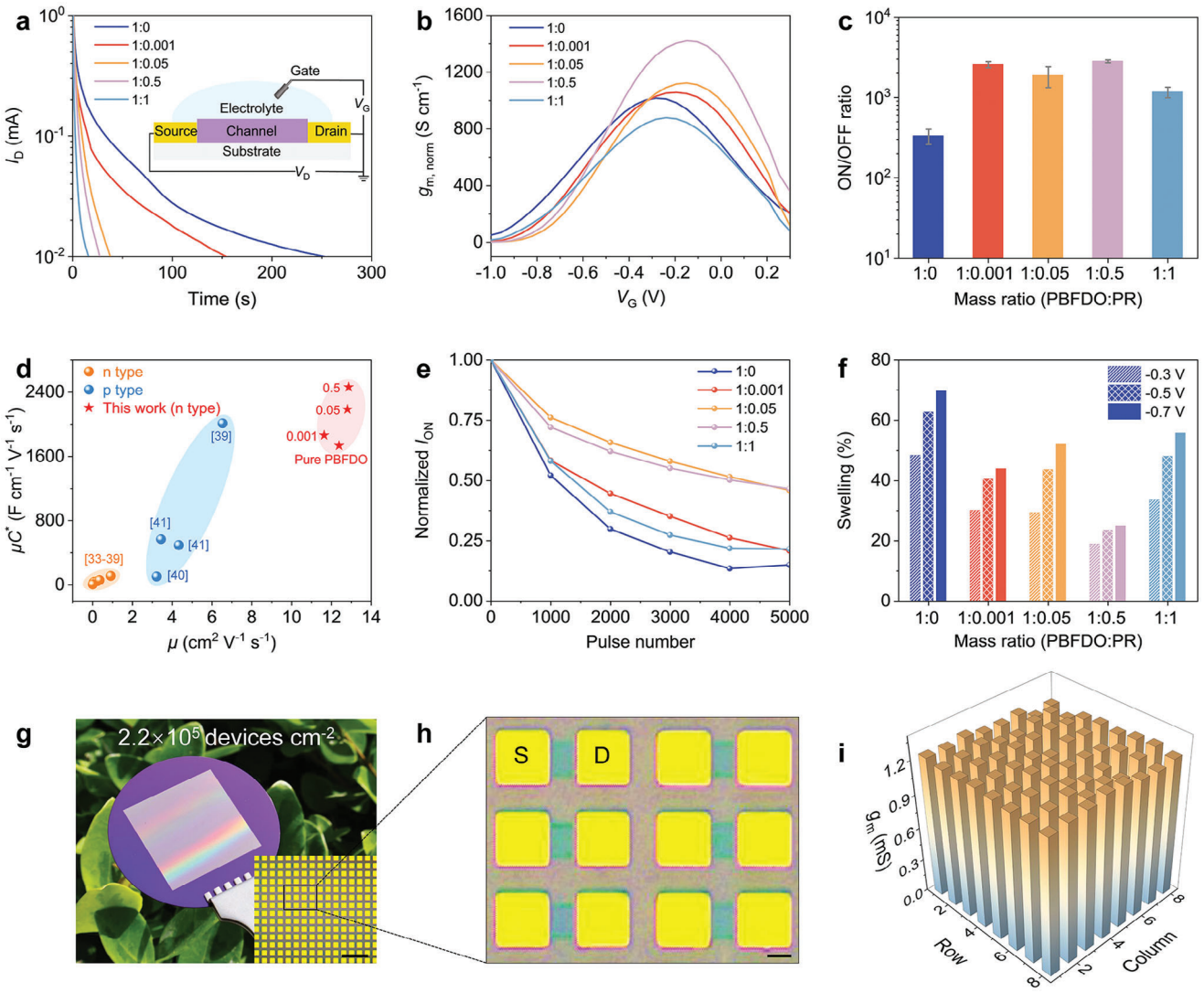


Figure 4. a) Initial dedoping process of OECTs based on pristine PBFDO film and PBFDO:PR films with different mass ratios. ($V_G = -1$ V, $V_D = 0.6$ V) b) Comparison of normalized transconductance values for pristine PBFDO and PBFDO:PR patterned OECTs ($V_D = 0.6$ V). c) Comparison of the ON/OFF ratio for pristine PBFDO and PBFDO:PR patterned OECTs. d) Comparison of μC^* and mobility of PBFDO: PR-based OECTs and previously reported OECTs.[36–44] e) Stability of pristine PBFDO and PBFDO:PR for patterned OECTs after 5000 voltage pulse cycles. f) Active swelling behavior of pristine PBFDO and different mass ratio PBFDO:PR films at various potentials. g) Image of a 2-inch wafer-scale OECT array composed of PBFDO:PR OECTs. Inset: optical microscope image of the OECT array (scale bar: 50 µm). h) Optical microscopy images of PBFDO:PR OECT arrays, indicating source and drain (scale bar: 5 µm). i) Transconductance (g_m) distribution of 8 × 8 PBFDO:PR OECTs in the array.

Table 1. Summary of OECTs based on the different mass ratios of PBFDO:PR.

Ratio	I_{ON} [mA]	I_{OFF} [µA]	I_{ON}/I_{OFF} [×10^3]	$g_{m,\ norm}$ [×10^3 S cm^{-1}]	V_{TH} [a)] [V]	μC^* [b)] [×10^3 F cm^{-1} V^{-1} s^{-1}]
1:0	3.62 ± 0.36	12.53 ± 3.57	0.33 ± 0.07	0.97 ± 0.07	−0.94 ± 0.04	1.73 ± 0.20
1:0.001	2.42 ± 0.18	1.54 ± 0.46	2.57 ± 0.21	1.02 ± 0.15	−0.88 ± 0.02	1.86 ± 0.29
1:0.05	3.71 ± 0.43	1.79 ± 0.07	1.87 ± 0.55	1.21 ± 0.12	−0.88 ± 0.04	2.18 ± 0.18
1:0.5	3.67 ± 0.33	1.31 ± 0.01	2.82 ± 0.12	1.33 ± 0.10	−0.81 ± 0.02	2.46 ± 0.17
1:1	2.11 ± 0.53	2.10 ± 0.09	1.17 ± 0.17	0.91 ± 0.14	−0.89 ± 0.05	1.51 ± 0.23

[a)] The variation trend of the threshold voltage (V_{TH}) is shown in Figure S24 (Supporting Information); [b)] The μC^* product was extracted from the linear region of $I_D^{1/2}$ versus gate voltage (V_G) curves according to the equation: $I_D^{1/2} = (V_{TH} - V_G) \times ((Wd/2L)\ \mu C^*)^{1/2}$ (as shown in Figure S25, Supporting Information), where I_D, W, d, and L are source-drain current, channel width, thickness and length, respectively. In this study, the dimensions were set to $W = L = 500$ µm, and $d = 30$–50 nm. To ensure accuracy, the thickness of the films was independently measured for each device. All reported parameters are based on measurements performed on more than 6 devices.

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PR to achieve efficient patterning of PBFDO at ultra-low loading levels (<0.1 wt%). By introducing multiple crosslinking sites and non-covalent bonding interactions, the loading level of the crosslinker is significantly reduced, minimizing the formation of defects in thin film. This approach enables the fabrication of feature sizes down to 600 nm. Furthermore, the dual crosslinking network effectively regulates the molecular order and crosslinking density of PBFDO chains, resulting in significant enhancements in ion conduction pathways and electrochemical stability. Consequently, notable improvements are observed in OECTs, including an approximately order-of-magnitude increase in ON/OFF ratio and a 42% enhancement in μC^* (2460 F cm^{-1} V^{-1} s^{-1}), along with better operational stability compared to devices using pristine PBFDO. High-density (2.2×10^5 devices cm^{-2}) wafer-scale arrays with excellent uniformity are also demonstrated. This strategy endows the supramolecular crosslinker with the ability to act as a patterning agent while being a performance enhancer, effectively addressing the critical challenges of balancing patternability and electronic properties in OSCs. Our findings pave the way for new applications of supramolecular chemistry in organic electronics, potentially influencing material designs for flexible electronics, bioelectronics, and next-generation wearable devices, where precise patterning and device efficiency are essential. Additionally, it provides a novel platform for fundamental studies on crosslinking mechanisms and intermolecular interactions in OSCs without compromising their intrinsic properties.

4. Experimental Section

Materials: All chemicals used for the synthesis were used as received without further purification unless otherwise mentioned. Detailed synthetic procedures for PR can be found in the Supporting Information.

Direct Microlithography: DML of PBFDO was performed on silicon substrates, which were cleaned and treated with UV ozone for 5 min before use. Solutions of PBFDO and PR in different mass ratios were prepared by adding a specified amount of PR to the PBFDO solution and then gradually diluting, with all mixtures thoroughly mixed before spin-coating. Photoinitiator 1173 was added to all PBFDO solutions, with a mass ratio of 1:2 to PR. The pristine PBFDO and PBFDO:PR films were obtained by spin-coating at 1000 rpm in a glove box, followed by evaporation of residual solvent in a vacuum. The PBFDO:PR films were then exposed using a SUSS MA6 mask aligner, followed by development in DMSO to obtain patterned films. After development, the films were annealed on the hot plate at 100 °C for 10 min.

Characterizations of Films: AFM measurements were performed using a Bruker Dimension Icon device with a Bruker AFM probe of spring constant k = 3 N m^{-1} in tapping mode. The C═C bond changes of PEGMA in the covalent crosslinking process were recorded using a Bruker Vertex 70 Fourier-transform infrared spectrometer, employing an attenuated total reflection (ATR) attachment to obtain spectra from 1500 to 1700 cm^{-1}. An Invia-Reflex Raman spectrometer was used to collect spectra in the 500–3200 cm^{-1} range with an excitation wavelength of 532 nm. GIWAXS data was obtained at 1W1A Diffuse X-ray Scattering Station, Beijing Synchrotron Radiation Facility (BSRF-1W1A). The monochromatic of the light source was 1.54 Å. The data was recorded by using the 2D image plate detector of EIGER 1M from Dectris, Switzerland. The total exposure time measured for all samples was 30 s.

Calculation and Simulation: Density functional theory (DFT) was used to calculate the binding energy of the H-bonds between various donors and acceptors in the blended film structures. Molecular dynamics (MD) was used to simulate the number of H-bonds forming between PBFDO and PR in the dual crosslinked network. More detailed information can be found in the Supporting Information.

Electrochemical Characterization: In situ UV–vis spectra were measured using a Lambda 950 UV–vis spectrophotometer with a wavelength range of 300–1600 nm. A two-electrode setup was employed, with an Ag/AgCl electrode used as the reference/counter electrode. The sample was placed in a cuvette and immersed in an ionic liquid electrolyte ($[EMIM]^+$ $[TFSI]^-$). A Keithley 2450 source meter applied voltage, with each voltage applied for 2 min before scanning the spectrum. EQCM-D measurements were performed with a Q-sense analyzer (QE401, Biolin Scientific) using gold-coated crystal sensors. Detailed test methods can be found in the Supporting Information.

Organic Electrochemical Transistor Fabrication and Characterization: All devices were fabricated on Si/100 nm SiO_2 wafers, with substrate processing steps the same as described previously. Subsequently, source and drain electrodes (3 nm Cr, 100 nm Au) on the substrate were deposited by thermal evaporation and patterned by photolithography using an UV Maskless Lithography machine (TuoTuo Technology, UV Litho-ACA). PBFDO:PR films were patterned by DML to define the channel, while the pristine PBFDO was wiped after spin coating to form the channel. Ag/AgCl pellets were used as the gate electrode for device testing, and $[EMIM]^+$ $[TFSI]^-$ was used as the electrolyte solution. Electrical characterization (output, transfer, and stability tests) of the OECTs was conducted at room temperature using two Keithley 2450 source meters and an electrical probe station. Film thicknesses were measured with a Bruker DektakXT stylus profilometer.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

J.Z., Y.Z., and H.J. contributed equally to this work. This research was supported by the National Key R&D Program of China (2023YFC3603500, 2024YFF0508603), the National Natural Science Foundation of China (52373282, 52173156, 52273316, 92372102, 22073094, 21474109), the Science and Technology Commission of Shanghai Municipality (23ZR1402000), the Fundamental Research Funds for the Central Universities (2232024Y-01, 223202023G-02, CUSF-DH-T-2024015, CUSF-DH-T-2024003), the Sichuan Science and Technology Program (2022NSFSC0877), the Science and Technology Development Program of Jilin Province (20240602003RC, 20210402059GH), the State Key Laboratory of Molecular Engineering of Polymers (Fudan University) (K2024-11), the Program for Young Scholars in Regional Development of CAS, the Central University Basic Research Business Funding Project (2023SCU12021). The authors acknowledged the cooperation of the Beijing Synchrotron Radiation Facility (BSRF-1W1A) and Shanghai Synchrotron Radiation Facility (SSRF) BL02U2 and BL16B1 for GIWAXS measurements.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

direct microlithography, organic electrochemical transistors, organic semiconductors, supramolecular photocrosslinker

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Received: November 11, 2024
Revised: January 7, 2025
Published online: February 4, 2025

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